Guided Soft Actor–Critic

Blind Go2-W Locomotion — Theory and Implementation

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Outline

The problem

Blind locomotion is a POMDP

In simulation we can read anything: body linear velocity, a height scan of the terrain, contact forces, friction, every randomized dynamics parameter.

On the real robot almost none of that exists. The deployed policy sees only proprioception: joint encoders, IMU, its own previous action.

The asymmetry

Training-time information $\gg$ deployment-time information. A policy trained on what it *cannot* have at deployment is useless; a policy that ignores what the simulator knows is wasteful.

Guided SAC is one answer: let a *privileged* network help train a *blind* network that is the only thing shipped.

Two observation channels

Proprioceptive, noisy, 5-frame history
5×57: base ang. vel. (3), projected gravity (3), velocity command (3), joint pos (16), joint vel (16), previous action (16)

Privileged, clean, single frame
$57 + 3 + 187$: same frame without noise, base *linear* velocity (3), height scan (187)

Action space: 16 dims = 12 leg position targets + 4 wheel velocity targets, tanh-squashed and scaled by per-dimension bounds (legs 5.0, wheels 10.0).

Guided SAC: the idea

Origin, and what we changed

Haklıdır & Temeltaş (2021) and Çerçi & Temeltaş (2024) train **two actors**: a *guiding actor* that sees clean state, and a *control actor* that sees degraded state and is pulled toward the guide's actions.

	Çerçi (2024)	**This project**
Why the control actor is handicapped	adversarial perturbation	*genuine* partial observability
Guiding actor sees	clean state s_t	privileged $\tilde{s}_t$ (height scan, lin. vel.)
Control actor sees	perturbed s_t^p	proprioception o_t + history
Shapes	identical	**different** (285 vs 532)
Agents	$k = 3, 5, 7$	$k = 1$ (single agent)

Two consequences: the paper's $s_t^p \to o_t$ mapping is *not* cosmetic (different dimensionality), and every multi-agent construction collapses at $k = 1$ — it must be re-derived, not copied.

Guided SAC is three mechanisms, not one

M1 — Asymmetric critic

$Q(\tilde{s}_t, a)$ sees everything, $\pi_f(a|o_t)$ does not

cost: *zero* extra networks
fixes credit assignment

M2 — Guide as behavior policy

the teacher also *acts* in the environment

cost: one extra actor
widens state visitation

M3 — Action-space imitation

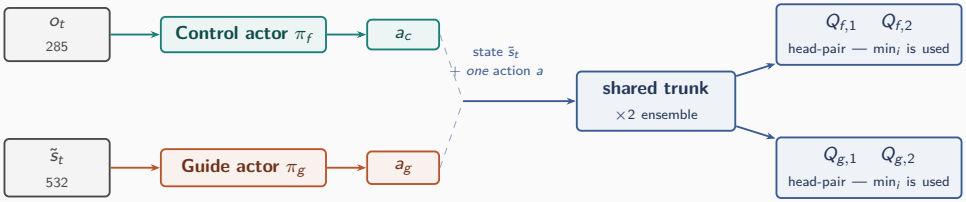
pull π_f toward π_g with weight λ

cost: one norm term
dense low-variance target

The literature ships these **bundled**. Separating them matters: M1 is free and not specific to Guided SAC, while M2 is the expensive one. Our design keeps M1, keeps M3 in softened form, and uses only a *small* amount of M2 — for a different reason than the original (next slides).

Architecture

Three networks, and two value functions



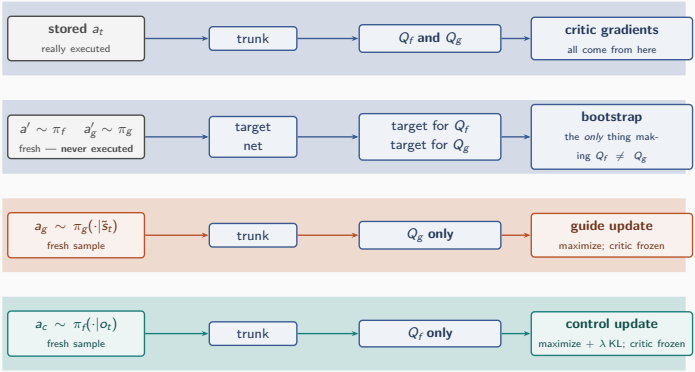
What the two heads mean

$Q_f(\tilde{s}_t, a)$ — how good is taking action a now, **if the blind student acts from then on**

$Q_g(\tilde{s}_t, a)$ — how good is taking action a now, **if the privileged teacher acts from then on**

Two trunks, and **two heads on each** — four in total. Within a pair the **smaller** value is always taken: that pessimism is the guide's only protection against its own value estimate running away.

Which action reaches which head

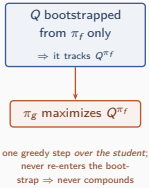


The critic's **gradients** come only from actions really executed (lane 1); the **bootstrap target** is evaluated at fresh samples that were never executed (lane 2) — exactly why the p_g mixture exists.

Why two value functions, and not one

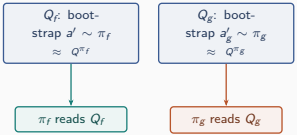
One head-pair per actor: π_f reads Q_f, π_g reads Q_g.

One shared critic (naive)



The teacher's advantage is **capped by construction**.

Two head-pairs (ours)



Each actor optimizes *its own* fixed point, so the teacher can run ahead — the student still optimizes the **deployed** policy's value.

Approximately: the n-step reward follows the **behaviour** trajectory, so Q_g estimates "take a_t, follow π_f for $n-1$ steps, *then* π_g" — an n-step-lagged Q^{π_g}.

Objectives

How the two value functions are trained

	Q_f **(student head)**	Q_g **(guide head)**
trained on	executed action a_t	executed action a_t
"and then?"	the student π_f continues	the teacher π_g continues
temperature used	α_f	α_g
so it estimates	$\approx Q^{\pi_f}$	$\approx Q^{\pi_g}$

Each head uses the ordinary SAC target — *the reward that really followed, plus the discounted value of the next state* — and only the policy used to value that next state differs. Distributional critic, n-step returns and the EMA target network are inherited unchanged.

Actors: one new term in the whole method

The guide is **plain SAC**. The student is **plain SAC plus one pull toward the guide** — that single term is the entire method.

Guide — ordinary SAC on privileged input, against its own heads:

$$\mathcal{L}_{\pi_g} \;=\; \mathbb{E}\Big[\, \alpha_g \log \pi_g(a_g|\tilde{s}_t) \;-\; \min_i Q_{g,i}(\tilde{s}_t, a_g) \,\Big]$$

Control — the same objective on proprioception, plus guidance:

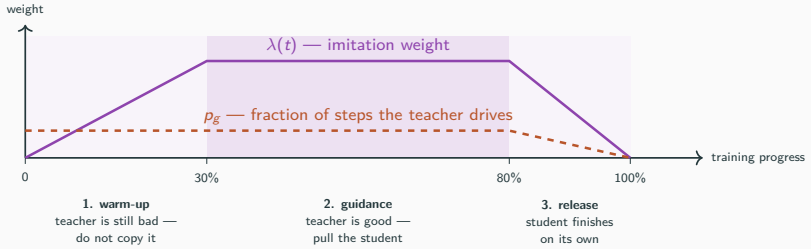
$$\mathcal{L}_{\pi_f} \;=\; \mathbb{E}\Big[\, \alpha_f \log \pi_f(a_c|o_t) \;-\; \min_i Q_{f,i}(\tilde{s}_t, a_c) \;+\; \underbrace{\lambda(t)\,\bar{q}\; D_{\mathrm{KL}}\big(\pi_f(\cdot|o_t) \,\big\|\, \mathrm{sg}[\pi_g(\cdot|\tilde{s}_t)]\big)}_{\text{the only term Guided SAC adds}} \,\Big]$$

- **KL, not L_2.** Both policies are tanh-Gaussian, so the KL is closed-form and matches the teacher's *variance*, not just its mean — the student may stay uncertain where the teacher is. The teacher is stop-gradded.
- $\bar{q} = \mathbb{E}|\min_i Q_{f,i}|$ keeps λ dimensionless as values grow; α_f, α_g are learned **separately**.

Schedules

λ: how hard the student is pulled toward the teacher

λ is the **weight on the imitation term**: $\lambda = 0$ means the student ignores the teacher entirely (plain SAC), larger λ pulls it harder toward copying the teacher. It is **not constant**:



p_g is a *different* knob — the small fraction of steps the teacher actually drives — so that Q_g is trained on actions the teacher really took.

The imitation gap — the method's real weakness

π_g solves the *fully observed* MDP. π_f must solve the *POMDP*. These are different problems, and projecting one onto the other is not optimal in general.

Concretely, for blind locomotion

A teacher that sees the height scan never learns to probe the ground with a foot, never steps cautiously into the unknown, never moves to enrich its own IMU signal — it has no need to. Imitating it can therefore *suppress exactly the behaviours that keep a blind robot alive*.

Mitigations built in, weakest to strongest:

1. **KL rather than L_2** — the student may stay uncertain where the teacher is.
2. $\lambda \to 0$ — guidance is removed before training ends.
3. **Latent supervision** (planned): supervise a learned *representation* $\hat{z} = E(o_t)$ against privileged z and let the policy decide how to use it — closes the gap by construction.

Deployment

Deployment — what survives

discarded: guide actor π_g • critic Q_f, Q_g • target critic • temperatures α_f, α_g • replay buffer • λ, p_g

- The deployed artifact is **architecturally identical** to a vanilla SAC actor. Guided training costs *nothing* at inference — the strongest practical argument for asymmetric methods.
- Nothing privileged can leak: the actor consumes a fixed 285-dim slice and the export path never reads the guide's files.

Implementation

Code map

Built as an **extension** of the FlashSAC backbone, never a fork: the vendored learning core is byte-identical, and without the `--guided` flag the pipeline is the unmodified baseline.

`gsac/config.py`	config dataclass; $\lambda(t)$, $p_g(t)$; fail-fast validation
`gsac/network.py`	`GuidedDoubleCritic`: shared trunks + second head-pair
`gsac/update.py`	the three losses above, each traced to its source equation
`gsac/agent.py`	`GuidedFlashSACAgent`: networks, rollout mixing, updates
`train.py --guided`	launcher flag; everything guided is behind it

The subclass overrides `sample_actions` (mixing), `update` (the three losses) and `save`/`load` (guide checkpoints); everything else is inherited. The teacher's action a_g is never stored — it is recomputed from the sampled batch at update time, so teacher/state alignment holds by construction.

One codebase, four experiments

Each rung switches on *one* mechanism, so any difference in outcome can be attributed to it:

	critic sees	**guide trained?**	**guide acts?**	**imitation**	
A1	o_t (blind)	no	no	no	vanilla SAC
A2	$\tilde{s}_t$ (privileged)	no	no	no	+ **M1**
A4	$\tilde{s}_t$	yes	$p_g > 0$	no ($\lambda = 0$)	+ **M2**
A5	$\tilde{s}_t$	yes	$p_g > 0$	yes ($\lambda > 0$)	+ **M3** = full Guided SAC

A1 → A2 is not "no teacher vs teacher" — neither has a second network. It is purely what the *critic* may see: vanilla SAC trains a blind critic too, while M1 hands it the height scan and leaves the actor blind. A teacher appears only at A4, and only *influences* the student at A5.

Which mechanism actually pays — and is the second actor worth its cost? The literature has never separated them.

Summary

- Blind locomotion is a genuine POMDP; the simulator knows far more than the robot ever will.
- Guided SAC bundles three separable mechanisms. We keep the free one (asymmetric critic), soften the risky one (imitation via scheduled KL), and use the expensive one sparingly and for a different reason than the original (value grounding).
- The single-agent reduction is not a copy: the teacher needs **its own value head**, otherwise its advantage is capped by construction.
- The imitation gap is the method's real theoretical weakness; latent supervision is the principled successor if action-space imitation does not pay.
- **Deployment cost is exactly zero** — the shipped network is a plain SAC actor on proprioception.